SUMMER EXAMINATION 2026

Artificial Intelligence - Version B

BSIT - 3rd Semester Time: 90 Minutes Max Marks: 80

Name: ______________________ Roll No.: ______________________ Date: ______________________

1. An expert system usually contains a knowledge base and an
(A) inference engine (B) image encoder
(C) order queue (D) optimizer

2. Q-learning estimates
(A) action values (B) class priors only
(C) parse probabilities (D) pixel edges

3. A confusion matrix summarizes
(A) classification outcomes (B) regression slopes
(C) search paths (D) cluster centroids

4. A ROC curve plots true positive rate against
(A) false positive rate (B) true negative rate
(C) loss (D) sample size

5. For TP=8 and FP=2, precision is
(A) 0.80 (B) 0.20
(C) 0.60 (D) 0.90

6. A policy maps states to
(A) actions or action probabilities (B) class labels only
(C) database rows (D) heuristic costs

7. Iterative deepening combines the low memory of DFS with the completeness of
(A) BFS (B) hill climbing
(C) minimax (D) beam search

8. A Markov decision process includes states, actions, transition model, rewards, and
(A) a discount factor (B) a compiler
(C) a parse tree (D) a domain vocabulary

9. Precision is the fraction of predicted positives that are
(A) actually positive (B) actually negative
(C) all observations (D) missed positives

10. A vector dot product measures alignment and is largest for vectors pointing
(A) in similar directions (B) in opposite directions
(C) at random (D) with zero length only

11. The F1 score is the harmonic mean of
(A) precision and recall (B) accuracy and loss
(C) bias and variance (D) speed and memory

12. Differential privacy protects individuals by adding controlled
(A) noise (B) labels
(C) search depth (D) network hops

13. If every edge costs 1, a path of depth 4 has cost
(A) 4 (B) 1
(C) 0 (D) 8

14. Backward chaining begins with
(A) a query goal (B) an empty database
(C) a sensor reading (D) a heuristic value

15. Weak AI is designed for
(A) a limited task (B) every intellectual task
(C) only robotics (D) only games

16. K-means assigns points to the nearest
(A) centroid (B) decision stump
(C) support vector (D) parse node

17. The Turing test evaluates whether a machine can
(A) converse indistinguishably from a human (B) sort numbers
(C) store facts (D) encrypt files

18. Naive Bayes assumes features are conditionally independent given the
(A) class (B) learning rate
(C) action (D) cluster index

19. Simulated annealing sometimes accepts
(A) a worse move (B) only a goal move
(C) no move (D) a duplicate state

20. A utility-based agent uses utility to
(A) rank desirable outcomes (B) tokenize text
(C) prune syntax trees (D) count database rows

21. Data leakage occurs when training uses information
(A) unavailable at prediction time (B) from a larger batch
(C) with missing values (D) after normalization

22. A constraint satisfaction problem consists of variables, domains, and
(A) constraints (B) rewards
(C) gradients (D) tokens

23. A learning rate that is too large may cause training to
(A) diverge (B) stop overfitting automatically
(C) produce labels (D) become deterministic

24. Exploration in reinforcement learning means
(A) trying uncertain actions (B) always choosing the best known action
(C) removing rewards (D) stopping training

25. Depth-first search uses a frontier organized as a
(A) stack (B) queue
(C) priority table (D) hash ring

26. A rational agent chooses an action that maximizes
(A) expected performance (B) memory usage
(C) program length (D) randomness

27. A validation set is used mainly for
(A) model selection and tuning (B) final unbiased reporting
(C) data collection (D) encryption

28. An AI agent's performance measure specifies
(A) how success is evaluated (B) how files are compressed
(C) how tokens are split (D) how keys are stored

29. Supervised learning uses
(A) labeled examples (B) no observations
(C) only rewards (D) only rules

30. Responsible AI emphasizes fairness, transparency, privacy, and
(A) accountability (B) unlimited surveillance
(C) opaque decisions (D) forced automation

31. AI is best described as the study of agents that perceive and act
(A) intelligent behavior (B) database normalization
(C) file compression (D) packet framing

32. In STRIPS, an action has preconditions and
(A) effects (B) pixels
(C) gradients (D) priors

33. Reinforcement learning learns from
(A) rewards and penalties (B) class labels only
(C) SQL joins (D) grammar rules

34. Gradient descent updates parameters using the
(A) negative loss gradient (B) positive labels only
(C) largest feature (D) random action

35. The utility of an outcome represents its
(A) desirability (B) token count
(C) search depth (D) probability only

36. A star search evaluates a node with
(A) f(n)=g(n)+h(n) (B) f(n)=g(n)-h(n)
(C) f(n)=h(n)/g(n) (D) f(n)=g(n)h(n) only

37. If a fair binary event has probability 0.5, its odds of success are
(A) 1 to 1 (B) 1 to 2
(C) 2 to 1 (D) 0 to 1

38. Knowledge engineering converts domain expertise into
(A) a knowledge base (B) a pixel matrix
(C) a routing table (D) a loss curve

39. Hill climbing can fail at a
(A) local maximum (B) root node only
(C) terminal compiler (D) balanced database

40. In alpha-beta pruning, alpha is the best value found for
(A) MAX so far (B) MIN so far
(C) the root heuristic (D) a random leaf

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1. Overfitting means a model fits training data well but
(A) generalizes poorly (B) has no parameters
(C) cannot train (D) has low variance always

2. A model-based agent maintains
(A) an internal state (B) only a lookup table
(C) a random seed (D) a utility invoice

3. A convolutional neural network is especially suited to
(A) spatial patterns (B) symbolic proofs only
(C) transaction logs only (D) shortest paths

4. Minimax assumes that the opponent
(A) acts optimally (B) acts randomly
(C) never moves (D) maximizes our score

5. A proposition is a statement that is
(A) true or false (B) always numerical
(C) a probability distribution (D) a neural layer

6. Pooling in a CNN commonly reduces
(A) spatial dimensions (B) the number of classes
(C) training labels (D) token vocabulary only

7. In PEAS, the E stands for
(A) Environment (B) Execution
(C) Encoding (D) Evaluation

8. A heuristic function estimates
(A) cost from a state to a goal (B) the exact training label
(C) network bandwidth (D) number of agents

9. PCA projects data onto directions of greatest
(A) variance (B) class count
(C) reward (D) text frequency only

10. Arc consistency removes values that
(A) have no supporting neighbor value (B) are globally optimal
(C) are duplicated (D) are outside the root

11. The Q-learning update uses the reward plus discounted
(A) next-state best Q value (B) current input size
(C) training accuracy (D) heuristic depth

12. High bias commonly causes
(A) underfitting (B) memorization
(C) data leakage (D) high variance only

13. A heuristic is admissible when it
(A) never overestimates true remaining cost (B) always equals true cost
(C) is negative (D) ignores the goal

14. Alpha-beta pruning changes minimax results
(A) never (B) only at even depth
(C) always (D) only with chance nodes

15. An LSTM uses gates to help preserve
(A) long-range dependencies (B) database keys
(C) pixel colors only (D) search depth

16. Forward chaining applies rules from
(A) known facts toward conclusions (B) goals toward facts
(C) random facts (D) only contradictions

17. Uniform-cost search selects the frontier path with
(A) lowest accumulated cost (B) lowest depth
(C) highest heuristic (D) most children

18. Backtracking search usually assigns
(A) one variable at a time (B) all variables randomly
(C) only goal variables (D) no values

19. Fuzzy logic allows truth values
(A) between 0 and 1 (B) only -1 and 1
(C) only 0 or 1 (D) above 100

20. An RNN is designed to process
(A) sequences (B) static tables only
(C) graphs only (D) images without order

21. A learning rate that is too small usually makes convergence
(A) slow (B) impossible in every case
(C) random (D) independent of data

22. A fully observable environment provides the agent with
(A) complete relevant state information (B) only rewards
(C) no sensors (D) past actions only

23. Breadth-first search expands the
(A) shallowest frontier node (B) lowest heuristic node
(C) deepest node (D) random node

24. The discount factor gamma is normally constrained to the interval
(A) 0 to 1 (B) 1 to 2
(C) -2 to -1 (D) 10 to 100

25. A Bayesian network represents dependencies using a
(A) directed acyclic graph (B) stack
(C) confusion matrix (D) parse tree only

26. An episodic task environment has actions that
(A) do not affect later episodes (B) always require planning
(C) must be continuous (D) have no observations

27. A false positive occurs when a negative case is
(A) predicted positive (B) predicted negative
(C) left unlabeled (D) counted twice

28. Recall is the fraction of actual positives that are
(A) correctly found (B) predicted negative
(C) all negatives (D) true negatives only

29. An agent's sensors provide percepts, while its actuators produce
(A) actions (B) heuristics
(C) constraints (D) utilities

30. The determinant of a 2 by 2 matrix [[a,b],[c,d]] is
(A) ad-bc (B) ab+cd
(C) ac-bd (D) a+b+c+d

31. Unsupervised learning seeks structure in
(A) unlabeled data (B) test labels
(C) utility functions (D) action policies

32. Backward planning works backward from
(A) the goal (B) the first percept
(C) the largest domain (D) the reward table

33. Forward planning starts at
(A) the initial state (B) the goal state
(C) the deepest leaf (D) the utility maximum

34. Cross-entropy penalizes a confident prediction that is
(A) wrong (B) correct
(C) uniform (D) missing a feature

35. A reflex agent selects actions mainly from
(A) the current percept (B) a learned value function
(C) a planning graph (D) a proof tree

36. A model card is intended to document
(A) model purpose, limits, and performance (B) network routes
(C) database indexes (D) source code licenses only

37. A perceptron computes a weighted sum followed by
(A) an activation (B) a database join
(C) a graph search (D) a token split

38. The principal purpose of normalization in ML is to
(A) put features on comparable scales (B) create labels
(C) increase leakage (D) remove all outliers

39. The MRV heuristic chooses the variable with
(A) fewest legal values (B) largest domain
(C) highest cost (D) most neighbors only

40. A goal test checks whether a state satisfies
(A) the goal condition (B) the heuristic estimate
(C) the action cost (D) the percept history